Granuloma Annulare

A localized ring of beaded papules on the extremities is typical; generalized, subcutaneous, perforating, and patch subtypes also occur.
The cause is unknown, and the pathogenesis is poorly understood.
Pathologic features consist of granulomatous inflammation in a palisaded or interstitial pattern associated with varying degrees of connective tissue degeneration and mucin deposition.
Most cases resolve spontaneously within 2 years.

Etiologic Associations

Photodamaged Skin

Granuloma annulare often occurs on photodamaged skin and is believed to represent a granulomatous reaction to actinic elastosis. Its relationship to granuloma annulare is debated.

Drugs

Granuloma annulare-like drug reactions are reported with gold therapy and treatment with allopurinol, diclofenac, quinidine, intranasal calcitonin, and amlodipine. An interstitial granulomatous drug reaction linked to angiotensin-converting enzyme inhibitors, calcium channel blockers, and other medications is considered a distinct entity but may clinically and histologically mimic granuloma annulare.

Diabetes Mellitus and Thyroid Disease

The development of granuloma annulare in patients with diabetes mellitus is well documented. The association appears stronger with type 1 insulin-dependent diabetes, although cases are also reported in type 2 non-insulin-dependent disease. Localized, generalized, subcutaneous nodular, and perforating forms of granuloma annulare have been observed in diabetic patients. Granuloma annulare rarely precedes the onset of diabetes.

The histopathologic similarity between granuloma annulare and necrobiosis lipoidica diabeticorum, along with the coexistence of both conditions in some diabetic individuals, supports a possible association. However, the majority of patients with granuloma annulare do not have diabetes mellitus, and studies attempting to establish a causal link have produced conflicting results.

Granuloma annulare has also been reported in patients with thyroiditis, hypothyroidism, and thyroid adenoma.

Malignancy

An association between granuloma annulare and malignancy has been reported, particularly in adult patients with Hodgkin and non-Hodgkin lymphoma—including mycosis fungoides, Lennert lymphoma, and B-cell and T-cell large cell lymphocytic leukemia. It is less commonly associated with solid tumors, especially breast cancer.

Cutaneous lymphoma and other hematologic malignancies may mimic granuloma annulare both clinically and histopathologically. Distinguishing true granuloma annulare with atypical lymphocytes from cutaneous lymphoma obscured by a granulomatous infiltrate can be challenging.

Pathogenetic Mechanisms

The mechanisms leading to foci of altered connective tissue surrounded by granulomatous inflammation remain unclear. Proposed theories include: 1. A primary degenerative process of connective tissue initiating granulomatous inflammation. 2. A lymphocyte-mediated immune reaction resulting in macrophage activation and cytokine-mediated degradation of connective tissue. 3. A subtle vasculitis or other microangiopathy causing tissue injury.

Clinical Findings

History

The typical history involves one or more papules that slowly enlarge centrifugally with central clearing, forming annular lesions.